Logistic Regression

Turning Farm Data into Yes/No Decisions

Simple idea. Powerful outcomes. Foundation of risk-based advisories.

AGRICULTURE AI

Why Logistic Regression Matters in Agriculture

Farm decisions are often binary classification problems, not quantity predictions.

Frost Damage

Will frost damage occur tonight?

Disease Detection

Is the crop diseased or healthy?

Irrigation Trigger

Should irrigation be activated now?

The Farming Connection

Logistic Regression supports risk-aware decisions by converting farm questions into actionable AI outputs.

Farm Question	AI Output
Will frost damage occur?	Yes / No
Is crop diseased?	Diseased / Healthy
Should pesticide be applied?	Apply / Don't apply

These binary decisions help farmers take timely protective actions.

Why It's Called "Regression"

Despite the name, Logistic Regression predicts **probability**, not continuous values.

- Probability is converted to Yes/No
- Output range is 0 to 1
- Not continuous yield prediction

0-1

Output Range

Probability scale

Logistic Regression in Simple Words

Step 1

Estimates the probability of an event happening

Step 2

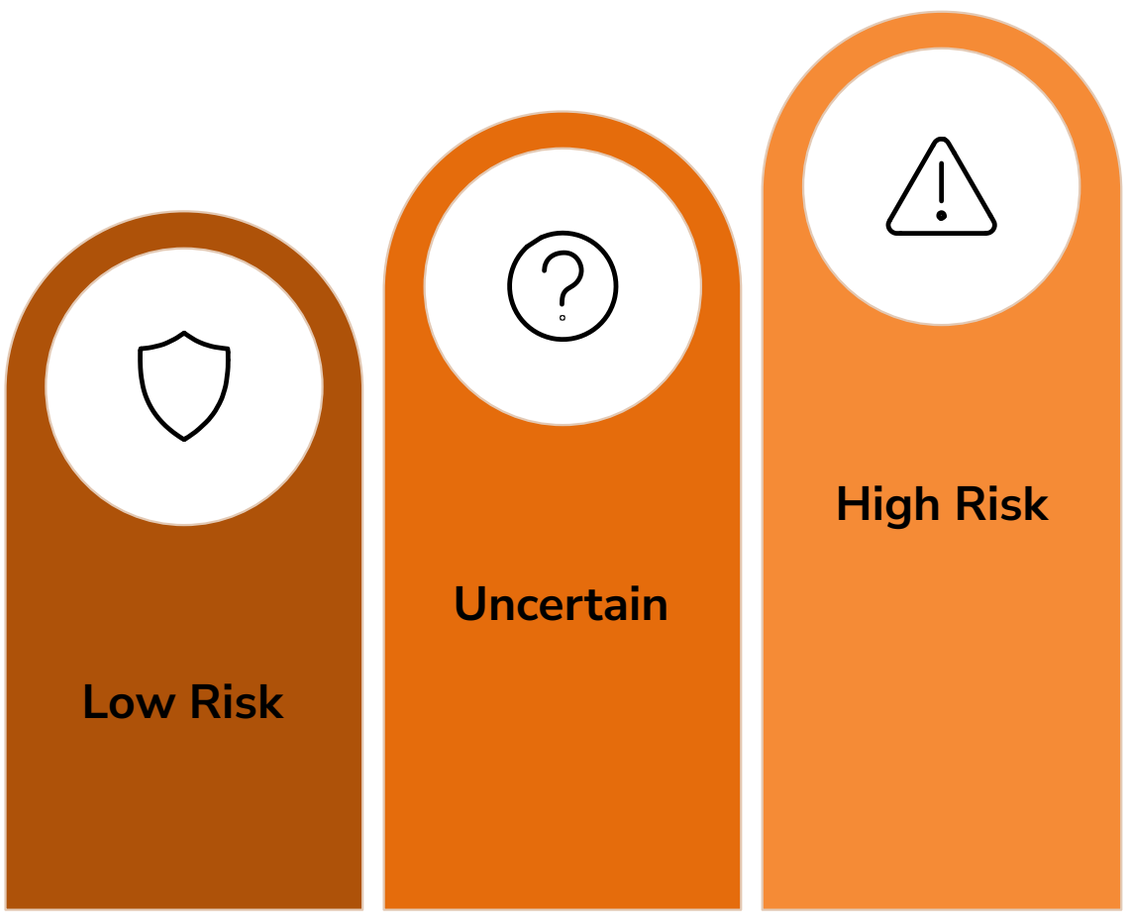
Applies a cutoff (e.g., 50%) to decide Yes or No

> "80% chance of frost damage" → Prepare protection.

CORE CONCEPT

The Sigmoid Function

The sigmoid curve is like a risk meter, not a hard switch. It converts any number into 0–1 probability with a smooth transition from No → Yes.



Sigmoid: Farming Intuition

Nature is gradual, not abrupt. The sigmoid function captures this reality.

Low Temperature

Low frost risk (near 0)

Very Low Temperature

High frost risk (near 1)

1 2 3

Borderline

Uncertain (0.4–0.6)

Example Farming Scenario

Inputs

- Minimum night temperature
- Humidity levels

Output

Frost damage risk (Yes/No)

Farmers need early warning, not exact numbers.

Sample Dataset

Each row represents one night's weather conditions. This Google Colab compatible dataset demonstrates frost risk classification.

```
import pandas as pd

# Frost risk dataset
data = {
    "Min_Temp_C": [2, 4, 6, 8, 1, 3, 7, 5],
    "Humidity": [90, 85, 70, 60, 95, 88, 65, 75],
    "Frost_Damage": [1, 1, 0, 0, 1, 1, 0, 0]
}

df = pd.DataFrame(data)
df
```

1 = Yes (Frost Damage), 0 = No (No Damage)

What the Data Means

01

Inputs

Weather conditions (temperature, humidity)

02

Output

Frost damage occurrence (Yes/No)

03

Learning Type

Labels make this supervised classification

Building the Logistic Regression Model

The model learns risk patterns from historical weather data and frost outcomes.

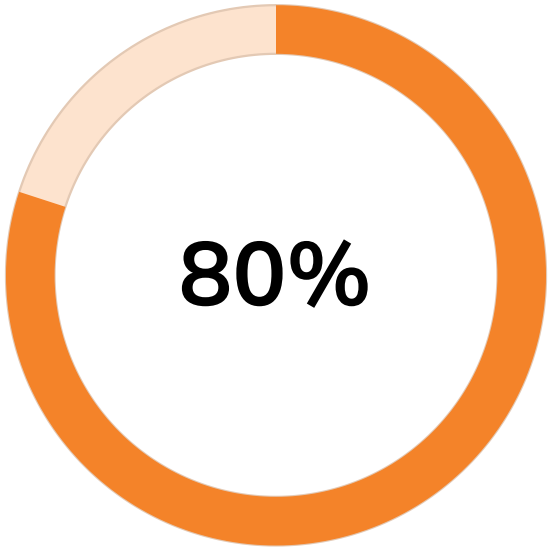
```
from sklearn.linear_model import LogisticRegression

X = df[["Min_Temp_C", "Humidity"]]  # Inputs
y = df["Frost_Damage"]  # Output (Yes/No)

model = LogisticRegression()
model.fit(X, y)
```

The model is now trained and ready to predict frost risk.

Predicting Probability

```
# Predict probability
model.predict_proba([[3, 90]])
```

Output: `[0.2, 0.8]`



Frost Damage Risk

Prepare frost protection measures

Converting Probability to Yes/No

The model uses a default 50% threshold to make binary decisions.

```
model.predict([[3, 90]])
```

≥ 0.5

Predicts Yes

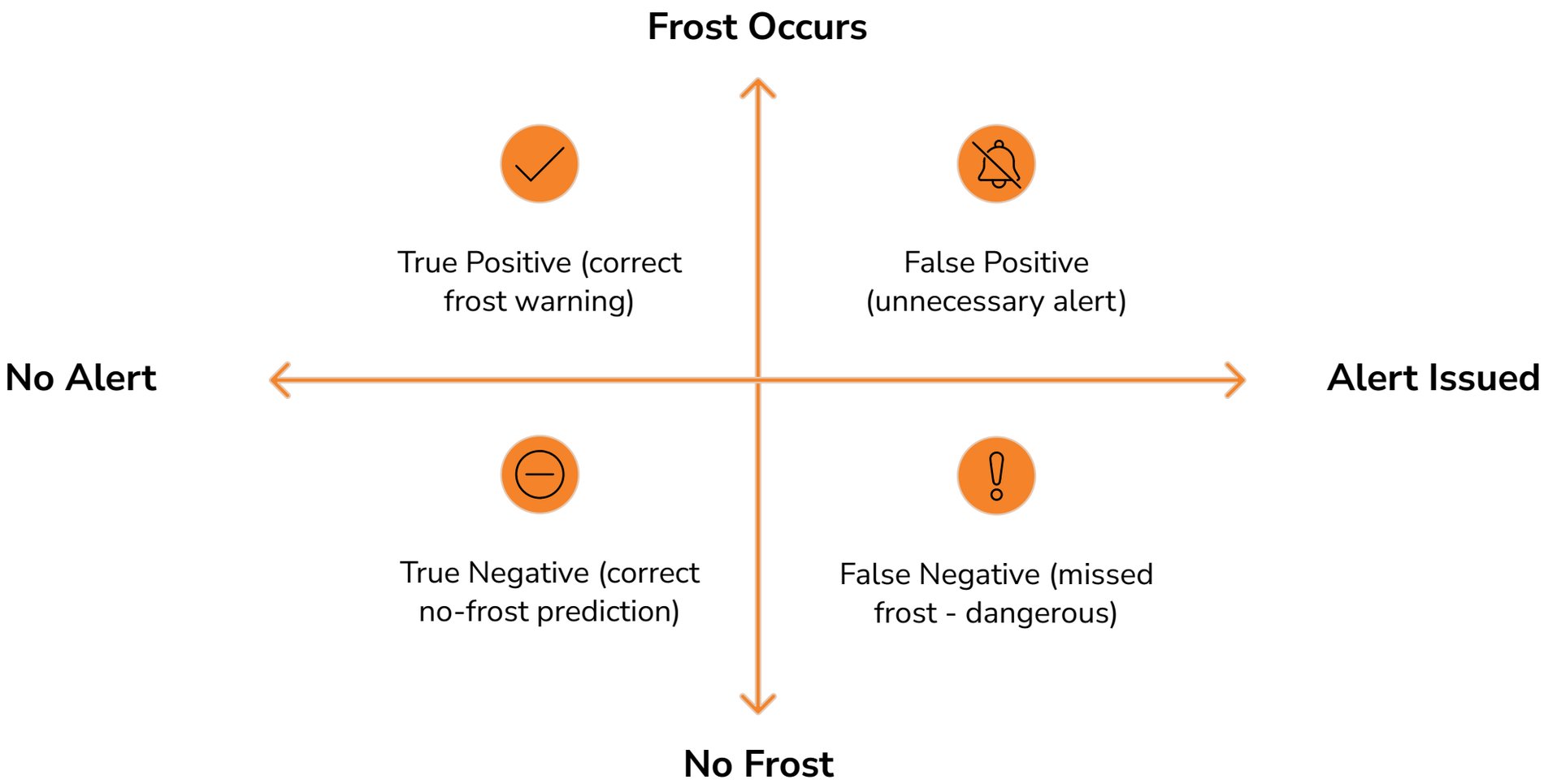
< 0.5

Predicts **No**

EVALUATION

Confusion Matrix: Why We Need It

Like evaluating an early-warning system, the confusion matrix shows correct warnings, missed warnings, and false alarms.



Confusion Matrix Demonstration

```
from sklearn.metrics import confusion_matrix

y_pred = model.predict(X)
confusion_matrix(y, y_pred)
```

True Positive

Correct frost warning

False Negative

Missed frost (dangerous!)

False Positive

Unnecessary alert

Accuracy: What It Means

```
from sklearn.metrics import accuracy_score

accuracy_score(y, y_pred)
```

Accuracy = % of correct decisions

87%

Model Accuracy

Correct frost decisions

Accuracy vs Reality

High accuracy alone is not enough. In agriculture, false negatives can be disastrous.

Rare Events

Disease and frost don't occur frequently

Costly Misses

Missing an event causes significant crop loss

Beyond Accuracy

Need to evaluate precision and recall

☆ CASE STUDY

Real Agriculture Application

Frost Advisory System

Data Collection

Logistic regression used on weather data

Risk Calculation

Output: Probability of frost damage

Alert System

Farmers alerted when risk > 70%

Results

Reduced crop loss, timely actions, higher trust

Motivation for Non-Programmers

You are not learning formulas. You are learning how to make better decisions under uncertainty.

Quantify Risk

How AI converts data into probability and confidence levels

Make Decisions

How Yes/No decisions are made from uncertain information

Question Advisories

How to evaluate and challenge unreliable recommendations

> Logistic regression builds confidence under uncertainty.

Key Takeaways

Core Concept

Probability-based Yes/No decisions

Sigmoid Function

Converts inputs to risk probability

Confusion Matrix

Evaluates warning quality

Accuracy Metric

Measures overall correctness

Essential for frost & disease alerts, risk-based advisories, and precision agriculture.

Predict risk early. Act wisely. Protect the harvest.